Dashboard & Visualization Design



SophiArch

Why Design Matters

- A dashboard is a **conversation**, not a data dump
- Readers decide in **~5 seconds** whether it is useful
- Good design = the reader finds the answer **without you in the room**
- Same data, better layout → faster, better decisions

> *If the reader has to ask "what am I looking at?", the design failed*

© 2026 SophiArch Data Visualisation course

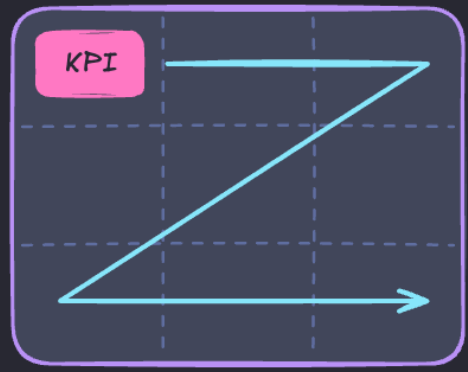
Start with Three Questions

- **Who** is reading?
  - Manager: big numbers, trends
  - Analyst: details, filters, tables
- **What** question must it answer?
  - e.g. "Are we on track against plan this quarter?"
- **What action** should they take after reading?

> *Write the answer in one sentence before you drag in a single visual*

© 2026 SophiArch Data Visualisation course

Layout Principles



© 2026 SophiArch Data Visualisation course

1. Placement — Most Important Top-Left

- Eyes start at the **upper-left**
- Then scan left → right, top → down
- Put the **key message / KPIs** there
- Details go lower down

> *Exception: right-to-left languages (Arabic/Hebrew)*

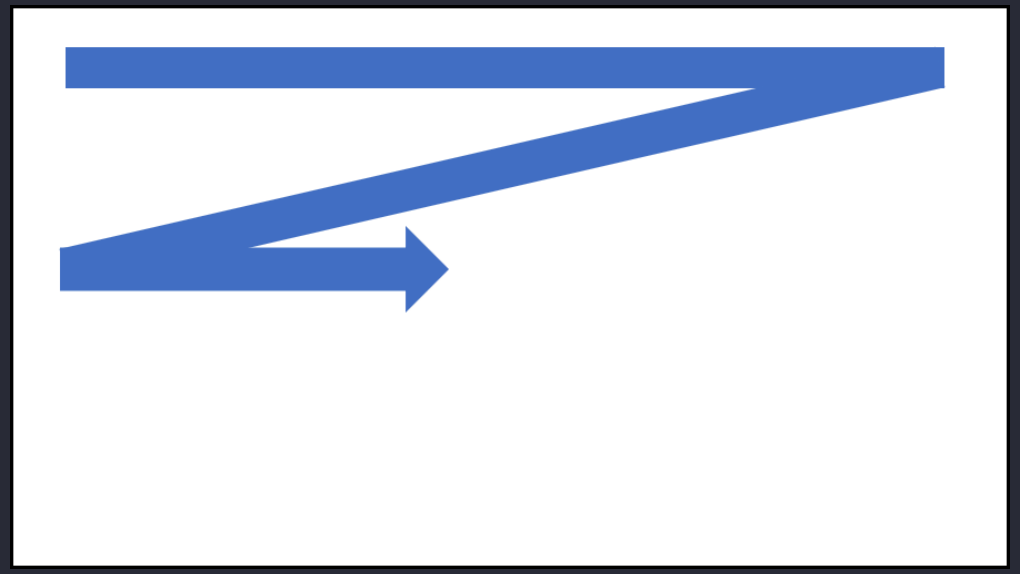
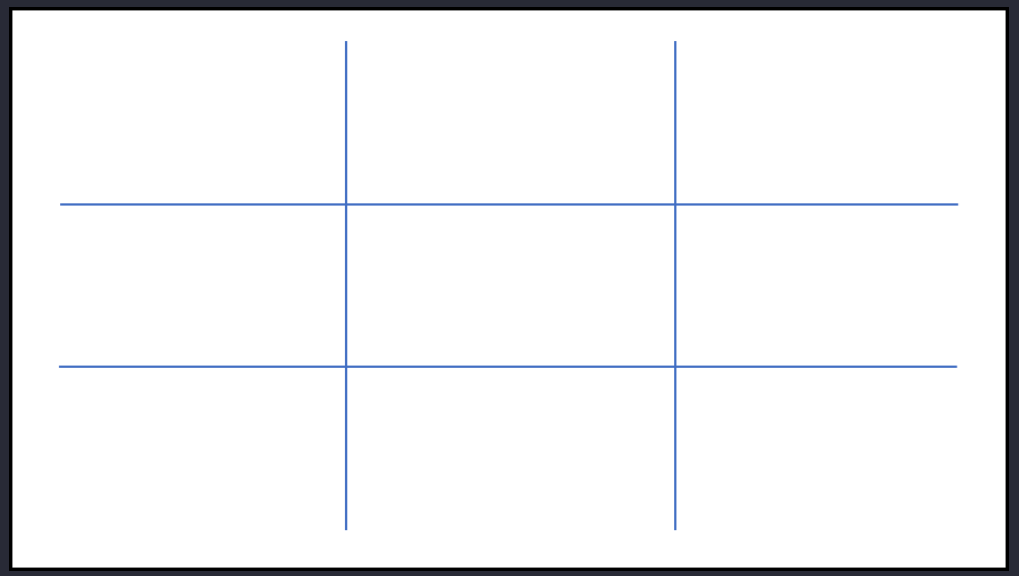


Image: Microsoft, *Design the analytical report layout*, Microsoft Learn

2. Placement — Rule of Thirds

- Split the page into a **3 × 3 grid**
- Place key visuals on the **lines / intersections**



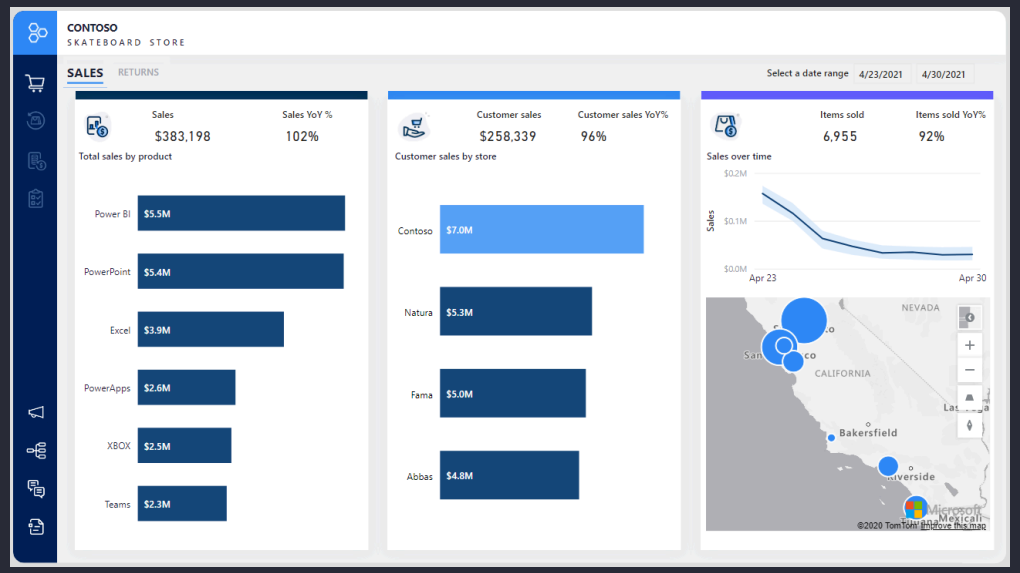


Image: Microsoft, *Design the analytical report layout*, Microsoft Learn

3. Balance — Golden Ratio

- Avoid one side "heavy", the other empty
- Golden ratio ≈ **1 : 1.618**
  - e.g. narrow panel + wide chart
- Big visuals carry more "weight"
  - spread them evenly

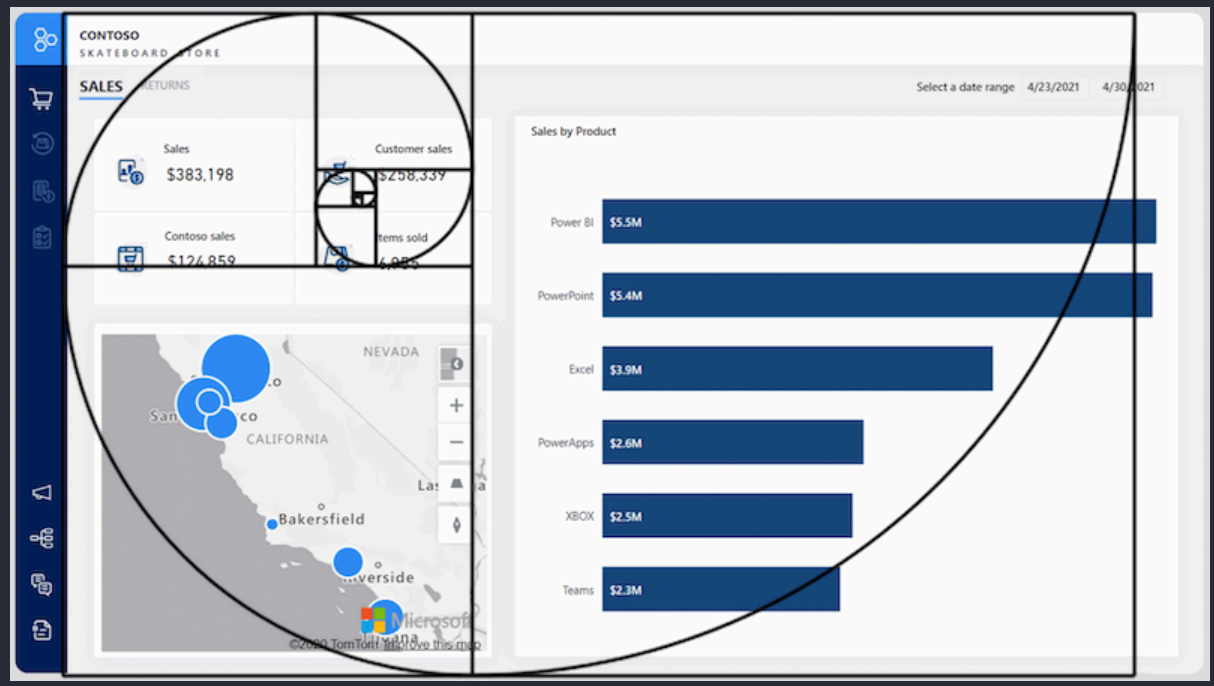


Image: Microsoft, *Design the analytical report layout*, Microsoft Learn

4. Proximity — Group Related Visuals

- Related visuals sit **near each other**
- Use **whitespace** to separate groups
- A shape / background box can frame a group



Image: Microsoft, *Design the analytical report layout*, Microsoft Learn

5. Contrast — Make the Important Stand Out

- Use contrasting **colour / size** to highlight
  - one key number, one bar, actual vs. target
- If everything is highlighted, **nothing** is



Image: Microsoft, *Design the analytical report layout*, Microsoft Learn

6. Repetition — Be Consistent

- Same **fonts, colours, title style, borders** everywhere
- Same colour = same meaning across pages
- Ties the report together visually

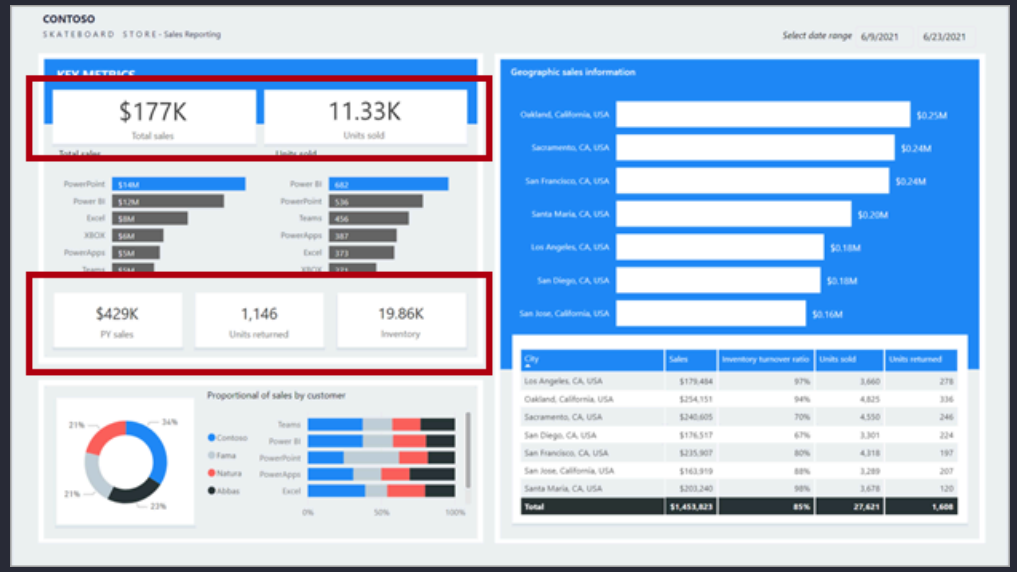


Image: Microsoft, *Design the analytical report layout*, Microsoft Learn

Dashboard Layout Ideas



© 2026 SophiArch Data Visualisation course

Sample Layout — Top-Down Story

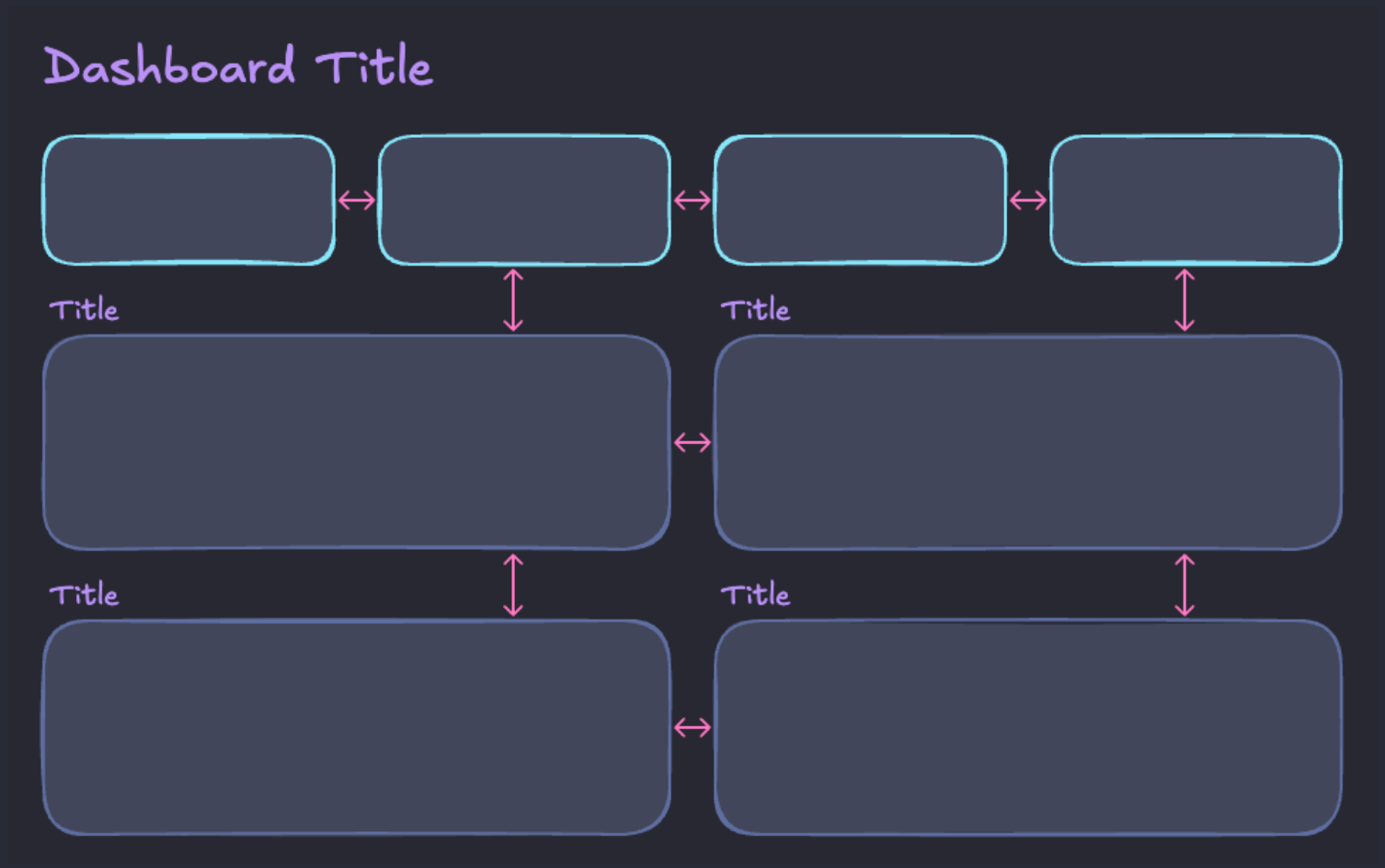
- **Header** — title, slicers, logo
- **KPI cards** — the headline numbers
- **High-level** — trends, summary charts
- **Detailed** — tables, breakdowns

> *Overview first, details on demand*



© 2026 SophiArch Data Visualisation course

Sample 2 × 2 Grid



© 2026 SophiArch Data Visualisation course

Sample 3 × 2 Grid



© 2026 SophiArch Data Visualisation course

Modified Grid — Real Examples



Image Source: Power BI Course Sample Dashboard / Analytics Mockup Template

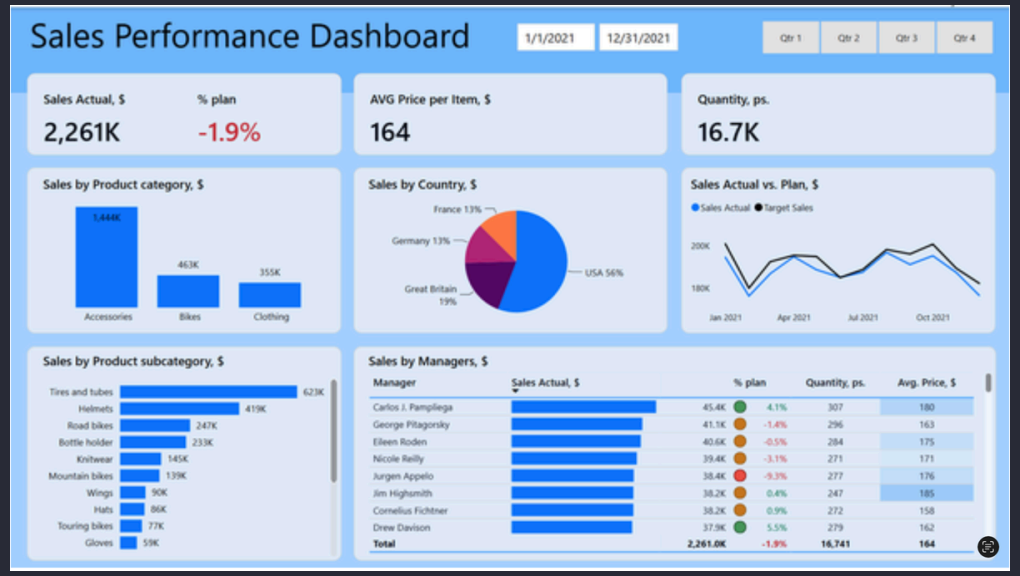


Image Source: Power BI Executive Sales Report Template (based on Microsoft AdventureWorks Sample Data)

- Start from a grid, then **merge cells** for bigger visuals

Choosing the Right Visual



© 2026 SophiArch Data Visualisation course

Match the Chart to the Question

I want to show...	Use	Power BI visual
One key number	Card / KPI	Card, KPI, Gauge (sparingly)
Compare categories	Bar / Column	Clustered bar / column
Trend over time	Line	Line, Area
Part of a whole (≤ 5 parts)	Stacked bar, Donut	100% stacked bar, Donut
Relationship between 2 measures	Scatter	Scatter chart
Exact values / detail	Table	Table, Matrix
Location	Map	Map, Filled map

© 2026 SophiArch Data Visualisation course

Design Effective KPI Cards

- Big **value**, small **label**
- Add **context**: vs. target, vs. last year
- Colour only the **variance** (▲ green / ▼ red)
- Keep **3–5 cards**, same size, one row



Image Source: Power BI Course Sample Dashboard / Analytics Mockup Template

> *"13,369" alone means nothing — "13,369 (–11% vs plan)" tells a story*

© 2026 SophiArch Data Visualisation course

Common Chart Mistakes

- **Pie with many slices** → use a sorted bar chart
- **3D charts** → distort size; always use 2D
- **Bar axis not starting at 0** → exaggerates differences
- **Too many series** on one line chart (> 4–5) → spaghetti
- **Unsorted bars** → reader hunts for the biggest
- **Table** when a chart would show the pattern faster

© 2026 SophiArch Data Visualisation course

Colour, Text & Clutter



© 2026 SophiArch Data Visualisation course

Use Colour with Purpose

- Pick **one theme**, max ~**3–5 colours**
- **Grey** for context, **one accent** for the focus
- Keep meaning consistent
  - green = good / red = bad, same category = same colour
- Don't rely on colour alone (~1 in 12 men are colour-blind)
  - add labels, icons or ± signs
- Check **contrast**: text must be readable on its background

© 2026 SophiArch Data Visualisation course

Titles, Labels & Text

- Every visual needs a **clear title**
  - Better: titles that state the **insight**
  - "Sales by Month" → "Sales peaked in November"
- Show **units**: $K, %, pcs
- Round numbers: **2.3M** not 2,261,437.58
- Data labels **or** axis — often not both
- Use at most **2 font sizes** for body text; title larger

© 2026 SophiArch Data Visualisation course

Declutter — Less is More

Remove

- Heavy gridlines & borders
- Background images & effects
- Duplicate legends / labels
- Visuals that don't answer the question

Keep

- Whitespace between groups
- Few visuals per page (**~5–7**)
- A legend only if needed
- One idea per visual

> *Every element should earn its place on the page*

© 2026 SophiArch Data Visualisation course

Power BI Tools That Help

Need	Where in Power BI Desktop
Align & equal spacing	**Format** tab → Align / Distribute
Snap to a grid	**View** tab → Gridlines, Snap to grid
Exact size & position	Format pane → General → Properties
Consistent styling	**Home** tab → Format painter; **View** tab → Themes
Find / layer / hide visuals	**View** tab → Selection pane
Control cross-filtering	**Format** tab → Edit interactions
Phone-friendly view	**View** tab → Mobile layout

© 2026 SophiArch Data Visualisation course

Slicers & Interactivity

- Keep slicers **together** — top bar or left panel
- **≤ 3–4 slicers** per page; more → use the Filter pane
- Show what's filtered (e.g. selected date range)
- Use **tooltips** for extra detail instead of extra visuals
- Use **drill-through** pages for the deep detail

© 2026 SophiArch Data Visualisation course

Navigation & Multi-Page Reports

- One page = **one topic** (Overview, Sales, Customers, ...)
- Same **header, slicers, nav buttons** in the same spot on every page
- Use a **page navigator** or buttons; meaningful page names
- Drill-through pages need a **Back** button
- **Hide** helper / tooltip pages before publishing
- Keep the default **16:9 canvas** (1280 × 720) unless you have a reason

© 2026 SophiArch Data Visualisation course

Before You Publish — Checklist

- □ Main message clear in **5 seconds**?
- □ KPIs in the **top-left / top row**?
- □ Related visuals **grouped**, gaps **equal**, edges **aligned**?
- □ Right **chart type**, sorted, axis from 0 for bars?
- □ **Titles, units**, rounded numbers?
- □ Limited, **consistent colours**; colour-blind safe?
- □ Slicers grouped, **interactions & navigation tested**?

© 2026 SophiArch Data Visualisation course

Key Takeaway

- **Purpose first** — who, what question, what action
- **Layout** — top-left KPIs, overview → detail, grid & whitespace
- **Visuals** — pick the chart for the question, keep it simple
- **Style** — few colours, clear titles, consistent everywhere

> *Make the reader's job easy*

© 2026 SophiArch Data Visualisation course

References

Topic	Source
Report design principles	Microsoft. (2024). *Design Power BI reports*. Microsoft Learn.
Layout images (slides 1–6)	Microsoft. (2025). *Design the analytical report layout*. Microsoft Learn. Images used for educational purposes.
Report design tips	Microsoft. (2024). *Tips for designing a great Power BI dashboard*. Microsoft Learn.
Accessibility	Microsoft. (2024). *Design Power BI reports for accessibility*. Microsoft Learn.
Visual types	Microsoft. (2024). *Visualization types in Power BI*. Microsoft Learn.
Example dashboards	Dashboards generated via Microsoft Power BI using custom analytics templates and standard sample dataset schemas (Microsoft AdventureWorks).
Data visualization	Knaflic, C. N. (2015). *Storytelling with Data*. Wiley.
Data-ink ratio	Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). Graphics Press.

© 2026 SophiArch Data Visualisation course

Thank You

- Questions?

© 2026 SophiArch Data Visualisation course